

JIANFEI CAO

CONTACT INFORMATION

Northeastern University
Department of Economics
College of Social Sciences and Humanities
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EDUCATION

2021	Ph.D. in Econometrics and Statistics, University of Chicago Booth School of Business
2015	Master of Science in Economics, University of Wisconsin - Madison
2013	Master of Financial Mathematics, University of Minnesota - Twin Cities
2011	Bachelor of Economics, Fudan University

EMPLOYMENT

2021- Assistant Professor, Department of Economics, Northeastern University

TEACHING AND RESEARCH FIELDS

Theoretical and Applied Econometrics
Specialized: Cluster-based inference, Dependent data, Panel Data, Machine Learning

PEER-REVIEWED JOURNAL ARTICLES

“[Synthetic Control Inference for Staggered Adoption](#),” with Shirley Lu and Hang Wu, forthcoming at *Econometrics Journal*, 2026.

“[Nephrologist Affiliation with Dialysis Facilities and Patient Trajectories in End Stage Renal Disease](#),” with Farbod Alinezhad, Gary J. Young, Richard Hirth, and Brady Post, *JAMA Network Open*, 2026, 9(4), e266156.

“[Robust IV Inference with Clustering Dependence](#),” *Econometrics Journal*, 2026, 29(1), 125–142.

“[Inference for Dependent Data with Learned Clusters](#),” with Christian Hansen, Damian Kozbur, and Lucciano Villacorta, *Review of Economics and Statistics*, 2025, 107(6), 1684–1701.

“[On the Empirical Content of the Beckerian Marriage Model](#),” with Xiaoxia Shi and Matthew Shum, *Economic Theory*, 2019, 67(2), 349–362.

BOOK CHAPTER

“[Principal Component and Static Factor Analysis](#),” with Chris Gu and Yike Wang, in *Macroeconomic Forecasting in the Era of Big Data* ed. by Peter Fuleky, 2020, 229–266. *Advanced Studies*

in *Theoretical and Applied Econometrics*, vol 52, Springer, Cham.

WORKING PAPERS

Revise and resubmit

“[Estimation and Inference for Synthetic Control Methods with Spillover Effects](#),” with Connor Dowd, available on arXiv, revision requested at *Econometric Reviews* on Apr 30, 2026.

“[Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data](#),” with Michael P. Leung, available on arXiv, revision requested at *Journal of Econometrics* on Apr 13, 2026.

“[Causal Diffusion Models for Counterfactual Outcome Distributions in Longitudinal Data](#)” with Farbod Alinezhad, Gary J. Young, and Brady Post, revision requested at *Annals of Applied Statistics* on Feb 27, 2026.

Submitted and under review

“[The Labor-Market Impact of Board Gender Quotas and Disclosure Policies](#),” with Yuling Han and Shirley Lu, submitted to *Journal of Economic Behavior & Organization* on May 5, 2026.

“[The Condition-Number Principle for Prototype Clustering](#),” with Romano Li, submitted to *Annals of Statistics* on Apr 8, 2026.

“[Assessing Heterogeneity of Treatment Effects](#),” with Tetsuya Kaji, available on arXiv, submitted to *Quantitative Economics* on Apr 2, 2025.

Working papers in preparation for submission

“Spectral Clustering for Network Cluster-Robust Inference,” with Romano Li.

WORK IN PROGRESS

“Calibrating LLM for Simulating Human Subjects,” with Shuo.

“Evaluating External Validity with Auxiliary Information,” with Ruonan Xu.

“Model Selection in Multiway Clustering”, with Hang Wu.

“Spillovers in Multiway Clustering,” with Tetsuya Kaji.

“Optimal Inference with Weak Identification,” with Tetsuya Kaji.

GRANT SUPPORT

\$29,100 in total

Northeastern CSSH Research Development Initiative

Title: “Calibrating LLM for Simulating Human Subjects”

Role: Principal Investigator

Dates: 8/15/2024-8/15/2025

Amount: \$7,500

Northeastern Economics Department Radivonyk Fund

Title: “Calibrating LLM for Simulating Human Subjects”

Role: Co-Principal Investigator (with Shuo Zhang)

Dates: 1/16/2024-1/16/2025

Amount: \$5,600

Northeastern Economics Department Radivonyk Fund

Title: “Synthetic Control Methods with Learned Spillover Structures”

Role: Principal Investigator

Dates: 5/2/2022-5/2/2023

Amount: \$3,000

Northeastern CSSH Multi-generational Research Teams Program

Title: “Synthetic Control Methods with Learned Spillover Structures”

Role: Principal Investigator

Dates: 9/1/2022-12/31/2022

Amount: \$13,000

ACADEMIC PRESENTATIONS

Invited seminars

2025

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **National Taiwan University**, Department of Economics, Taipei, 10/30/2025

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **Toronto Metropolitan University**, Department of Economics, Toronto, 4/1/2025

2024

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **City University of Hong Kong**, College of Business, Hong Kong, 12/16/2024

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **The Chinese University of Hong Kong-Shenzhen**, School of Management and Economics, Shenzhen, 11/21/2024

“Assessing Heterogeneity of Treatment Effects,” **University of Notre Dame**, Department of Economics, Notre Dame, 11/7/2024

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **Shanghai University of Finance and Economics**, School of Economics, Shanghai, 9/30/2024

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **Fudan University**, School of Management, Shanghai, 9/29/2024

2023

“Assessing Heterogeneity of Treatment Effects,” **Shandong University**, School of Economics, Jinan, 12/28/2023

“Assessing Heterogeneity of Treatment Effects,” **Fudan University**, School of Management, Shanghai, 12/27/2023

“Assessing Heterogeneity of Treatment Effects,” **University at Albany**, Department of Economics, Albany, 10/16/2023

2021

“Inference for Dependent Data with Learned Clusters,” **University of Naples Federico II**, Centre for Studies in Economics and Finance, remote, 5/18/2021

“Robust IV Inference with Clustering Dependence,” **Northeastern University**, Department of Economics, remote, 2/3/2021

“Robust IV Inference with Clustering Dependence,” **Erasmus University Rotterdam**, Econometric Institute; remote, 1/29/2021

“Robust IV Inference with Clustering Dependence,” **Washington State University**, Department of Economics, remote, 1/26/2021

“Robust IV Inference with Clustering Dependence,” **University of Connecticut**, Department of Economics, remote, 1/25/2021

“Robust IV Inference with Clustering Dependence,” **University of Georgia**, Department of Economics, remote, 1/22/2021

“Robust IV Inference with Clustering Dependence,” **National University of Singapore**, Department of Economics, remote, 1/19/2021

“Robust IV Inference with Clustering Dependence,” **University of Glasgow**, Adam Smith Business School, remote, 1/15/2021

“Robust IV Inference with Clustering Dependence,” **New Economic School**, remote, 1/13/2021

“Robust IV Inference with Clustering Dependence,” **Chinese University of Hong Kong**, Business school, remote, 1/12/2021

“Robust IV Inference with Clustering Dependence,” **Luohan Academy**, remote, 1/11/2021

“Robust IV Inference with Clustering Dependence,” **Peking University**, National School of Development, remote, 1/5/2021

Conference sessions

2026

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **COMP-STAT 2026**, Athens, Aug 2026 (scheduled)

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **Econometric Society Asia Meeting, East and Southeast Asia**, Hanoi, 8/1/2026 (scheduled)

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **Chinese Economists Society China Annual Conference**, Chengdu, Jul 2026 (scheduled)

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **MLiFE2026, Machine Learning in Finance and Econometrics**, Chengdu, Jun 2026 (scheduled)

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **Econometric Society Asia Meeting, China**, Hong Kong, 6/21/2026 (scheduled)

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **ESIF Economics and AI+ML Meeting** (hosted by Cornell University), Ithaca, 6/17/2026 (scheduled)

“Neighborhood Stability in Double/Debiased Machine Learning with Dependent Data,” **Chinese Economists Society North America Annual Conference** (hosted by Emory University), Atlanta, 3/28/2026

2025

“Assessing Heterogeneity of Treatment Effects,” **North American Winter Meeting of the Econometric Society**, San Francisco, 1/4/2025

2024

“Block Cross Fitting and Validation for Spatial Data,” **Greenline Workshop in Econometrics** (hosted by Boston College), Boston, 12/6/2024

“Clustered Cross-Validation and Cross-Fitting for Dependent Data,” **CEME Young Econometricians Conference** (hosted by Duke University), Durham, 9/14/2024

2023

“Assessing Heterogeneity of Treatment Effects,” **European Winter Meeting of the Econometric Society** (hosted by University of Manchester), Manchester, 12/18/2023

“Assessing Heterogeneity of Treatment Effects,” **Greenline Workshop in Econometrics** (hosted by Boston University), Boston, 12/1/2023

“Assessing Heterogeneity of Treatment Effects,” **Midwest Econometrics Group** (hosted by Federal Reserve Bank of Cleveland), Cleveland, 10/13/2023

“Assessing Heterogeneity of Treatment Effects,” **CEMP Workshop on Econometric Theory and Applications** (hosted by Jinan University), Guangzhou, 7/4/2023

“Estimation and Inference for Synthetic Control Methods with Spillover Effects,” **ENAR Spring Meeting**, Nashville, 3/20/2023

2020

“Robust IV Inference with Clustering Dependence,” **European Winter Meeting of the Econometric Society** (hosted by University of Nottingham), remote, 12/14/2020

“Inference for Dependent Data with Cluster Learning,” **World Congress of the Econometric Society** (hosted by Bocconi University), remote, 7/14/2020

“Inference for Dependent Data with Cluster Learning,” **North American Winter Meeting of the Econometric Society**, San Diego, 1/4/2020

2019

“Inference for Dependent Data with Cluster Learning,” **European Winter Meeting of the Econometric Society** (hosted by Erasmus University Rotterdam), Rotterdam, 12/16/2019

“Inference for Dependent Data with Cluster Learning,” **Midwest Econometrics Group** (hosted by Ohio State University), Columbus, 10/11/2019

“Estimation and Inference for Synthetic Control Methods with Spillover Effects,” **North America Summer Meeting of the Econometric Society** (hosted by University of Washington), Seattle, 6/30/2019

“Estimation and Inference for Synthetic Control Methods with Spillover Effects,” **China Meeting of the Econometric Society** (hosted by Jinan University), Guangzhou, 6/20/2019

TEACHING EXPERIENCE AT NORTHEASTERN

Courses			
Semester	Course#	Course Title	# of Students
Spring 2026	ECON 2560-09	Applied Econometrics	17/26
Spring 2026	ECON 2560-12	Applied Econometrics	23/35
Spring 2026	ECON 5210-01	Machine Learning	11/25
Spring 2026	ECON 5210-02	Machine Learning	10/25
Spring 2026	ECON 6140	Advanced Applied Econometrics	9/20
Spring 2025	ECON 2560-12	Applied Econometrics	31/35
Spring 2025	ECON 5200-03	Machine Learning and AI	22/22
Spring 2025	ECON 6140	Advanced Applied Econometrics	17/25
Spring 2025	ECON 7740	Applied Econometrics 2	11/15
Spring 2024	ECON 2560-07	Applied Econometrics	23/32
Spring 2024	ECON 5200-03	Machine Learning and AI	19/20
Spring 2024	ECON 6140	Advanced Applied Econometrics	19/20
Fall 2023	ECON 7740	Applied Econometrics 2	5/12
Spring 2023	ECON 2560-04	Applied Econometrics	11/28
Spring 2023	ECON 6140	Advanced Applied Econometrics	8/20
Fall 2022	ECON 2560-02	Applied Econometrics	22/30
Spring 2022	ECON 6140	Advanced Applied Econometrics	11/25

Doctoral students supervised

Ziyao Wang, Department of Economics (ABD, Ph.D. expected 2026). Three Essays in Empirical IO: Technological Change and Productivity. Committee member.

Farbod Alinezhad, Bouvé College of Health Sciences (Ph.D. 2025). The Determinants and Results of Vertical Integration of Hospitals and Physicians and Methods on the Use of Diffusion Models for Causal Inference. Committee member.

Jiacheng Liang, Department of Economics (Ph.D. 2023). Essays in Empirical Macroeconomics and Finance. Committee member.

SERVICE TO THE PROFESSION

Refereeing: *Annals of Applied Statistics*, *Econometric Reviews*, *Economics of Innovation and New Technology*, *Journal of Applied Econometrics*, *Journal of Econometrics*, *Journal of Political Economy*, *Microeconomics*, *Journal of Risk and Financial Management*, *Labour Economics*, *Review of Economics and Statistics*